V1 neural-field coefficients: sources and derivation

This note accompanies the V1 notebook and extends the model in `final_report.pdf`, confirmed by the author as similar to the supplied report. The gains below are reproducible phenomenological calibrations, not estimates of physiological synaptic strengths.

1 What is fixed by evidence, and what is assumed?

Order the populations as (E, PV, SST, VIP), with target rows and source columns. The four-population sign/zero mask and the no-VIP restriction are

$$\mathcal{M}_4 = \begin{pmatrix} + & - & - & 0 \\ + & - & - & 0 \\ + & 0 & 0 & - \\ + & 0 & - & 0 \end{pmatrix}, \qquad \mathcal{M}_3 = (\mathcal{M}_4)_{1:3,1:3}. \tag{1}$$

Every existing connection and its sign are preserved. Exact zeros represent the chosen reduction, not proof that the corresponding biological pathways never occur.

Pfeffer et al. [1]. The measured inhibitory motif motivates PV inhibition of E/PV, SST inhibition of the other retained populations, and preferential VIP inhibition of SST. It does not fix our effective gain magnitudes.

Billeh et al. [2]. Their Gaussian distance rule supplies the scales $s_{E\to E} = 114$, $s_{E\to PV} = 92$, $s_{E\to SST} = 103$, $s_{PV\to E} = 95$, $s_{SST\to E} = 85$, and $s_{PV\to PV} \simeq 120\,\mu$m. Their substitutions for other inhibitory pairs and Htr3a connections motivate our remaining widths; Htr3a is broader than VIP. Their feedforward model motivates driving E/PV, although equal E/PV drive is our assumption.

Campagnola et al. [3]. Cell-class/layer dependence of connection probability, strength and dynamics limits the interpretation of a single coarse matrix. No numerical gain is copied from this study.

Hahn et al. [4]. Their time-constant sensitivity analysis (Fig. 5E–H) motivates testing frequency sensitivity. It does *not* establish our exact time constants. No cited experiment uniquely determines those values.

1.1 From spatial measurements to field coefficients

For a unit-mass Gaussian, one spatial unit is $100\,\mu$m, and

$$P_{ij}(r) = C_{ij}\mathrm{e}^{-r^2/s_{ij}^2}, \qquad \Sigma_{ij} = \frac{(s_{ij}/100)^2}{2}, \qquad \widehat{W}_{ij}(q;p) = J_{ij}(p)\mathrm{e}^{-\Sigma_{ij}q^2/2}. \tag{2}$$

Thus J_{ij} is an integrated gain, not a probability or Gaussian peak height. The disk-average convention in [2], with $R = 75\,\mu$m, gives

$$P_{\mathrm{rep},ij} = \frac{1}{\pi R^2}\int_0^R 2\pi r C_{ij}\mathrm{e}^{-r^2/s_{ij}^2}\,dr = \frac{C_{ij}s_{ij}^2}{R^2}(1-\mathrm{e}^{-R^2/s_{ij}^2}). \tag{3}$$

Consequently $C_{ij} = P_{\mathrm{rep},ij}R^2/[s_{ij}^2(1-\mathrm{e}^{-R^2/s_{ij}^2})]$. If source density and effective per-connection strength were known, a planar reduction would give $J_{ij} \propto \rho_j w_{ij}\pi s_{ij}^2 C_{ij}$, with further activity-unit scaling. The legacy probability array does not supply these missing factors. The 1D marginal of the unit-mass 2D kernel preserves J_{ij}.

2 Complete coefficient specification

The source-nonlinearity convention is

$$\tau_i \partial_t u_i = -u_i + \sum_j W_{ij} * f_j(u_j) + I_i, \qquad f_j(u) = \ell(g_j(u-\theta_j)) - \ell(-g_j\theta_j), \quad \ell(z) = \frac{1}{1+\mathrm{e}^{-z}}. \tag{4}$$

Here u is a signed deviation variable, time is in ms, and $L = 47.6023$ model units $= 4.76023$ mm. Retain the legacy activation parameters and effective time scales:

Population	τ (ms)	g	θ
E	10	3.47682761	0.28483799
PV	12	7.34413013	0.12068148
SST	35	3.74650527	0.40137356
VIP	20	6	0.7

A one-pole filter has $h_i(t) = \tau_i^{-1}\mathrm{e}^{-t/\tau_i}$ for $t \geq 0$, so $\int_0^\infty t h_i(t) dt = \tau_i$. No measured population impulse responses were available to identify these moments. We therefore retain the ratios $1 : 1.2 : 3.5 : 2$, with a 10 ms reference, as explicit hypotheses. They are not measured membrane or synaptic-decay constants. Holding them and g, θ fixed isolates the change in connectivity; their decimal precision is not experimental precision.

The width matrix, including dynamically irrelevant placeholders on zero edges, is

$$s = \begin{pmatrix} 114 & 95 & 85 & 85 \\ 92 & 120 & 120 & 120 \\ 103 & 120 & 120 & 120 \\ 103 & 120 & 120 & 120 \end{pmatrix} \mu\mathrm{m}, \qquad \Sigma = (s/100)^2/2 \quad \text{entrywise.} \tag{5}$$

The $120\,\mu$m SST–VIP/SST–PV and $103\,\mu$m E–VIP scales are extrapolations in the reduction, not VIP-specific measurements.

Set $J(p) = (0.5 + p)G$, equivalently $A = 0.5G$ and $A_p = G$. The calibrated gains are

$$G_3 = \begin{pmatrix} 5.87511 & -1.36667 & -6.36306 \\ 4.74411 & -0.99825 & -8.11851 \\ 2.64504 & 0 & 0 \end{pmatrix}, \tag{6}$$

$$G_4 = \begin{pmatrix} 6.13537 & -1.42384 & -6.23105 & 0 \\ 5.13746 & -0.89269 & -7.81025 & 0 \\ 2.54421 & 0 & 0 & -16.04588 \\ 1.65365 & 0 & -3.21101 & 0 \end{pmatrix}. \tag{7}$$

The notebook uses the full-precision values in `simulations/output/v1_extension/calibration.json`. The printed matrices are rounded. The no-VIP block is calibrated separately; comparing these matched models is not an isolated causal VIP-ablation experiment.

3 Reproducible derivation of the new gains

The legacy full-scale matrix is

$$G_{\text{old}} = \begin{pmatrix} 6.22009 & -1.45455 & -6.48944 & 0 \\ 5.09798 & -0.90615 & -8.04066 & 0 \\ 2.67174 & 0 & 0 & -16 \\ 1.68844 & 0 & -3.13584 & 0 \end{pmatrix}. \tag{8}$$

Its original optimization history cannot be reconstructed. Instead, take this matrix (or its principal 3×3 block) as a prior and solve

$$\min_{-2 \le z_{ij} \le 2} \sum_{G_{\text{old},ij} \neq 0} z_{ij}^2, \qquad G_{ij} = G_{\text{old},ij} \mathrm{e}^{z_{ij}} \quad \text{on retained edges}, \tag{9}$$

with zeros fixed, subject to the four spectral targets

$$\begin{aligned} &\operatorname{Re} \lambda_d(q_0, p_0) = 0, && \partial_q \operatorname{Re} \lambda_d(q_0, p_0) = 0, \\ &\operatorname{Im} \lambda_d(q_0, p_0) = 0.08, && \max \operatorname{Re} \sigma(M(0, p_0)) = -0.001, \quad q_0 = 6\pi/L, \ p_0 = 0.25. \end{aligned} \tag{10}$$

The matrix M is derived in Report 2. These are design targets: mode 3, frequency 12.7324 Hz and a damped homogeneous mode, not measured V1 values. SLSQP starts at $z = 0$, uses tolerance 10^{-12} and a central q-difference of 10^{-4}. It finds a local constrained optimum; uniqueness/global optimality is not claimed. Numerical checks find the first crossing at $(p_0, k_0) = (0.25, 3)$, verify the square-lattice shells and a stable Gaussian tail, and obtain negative real cubic saturation coefficients. Unretuned VIP deletion instead gives a homogeneous Hopf at $p \simeq 0.153726$.

Forcing. At $p = p_0 - 0.0005$, drive E/PV equally at $2\omega_0$ and/or spatial mode 6. A real cosine of peak d has complex harmonic $d/2$. Solving for leading-order amplitude-Jacobian growth 10^{-4}/ms gives

Model	Temporal peak	Spatial peak	Each combined peak
3 populations	0.0022181	0.0061721	0.0016479
4 populations	0.0021273	0.0028546	0.0012339

Direct simulations check the resulting patterns separately from this asymptotic prediction. With gains fixed, ±20% changes in E/PV/SST time constants often remove the finite-wave-first scenario: these are conditional existence examples, not a robust physiological fit.

References

[1] C. K. Pfeffer et al. (2013), Inhibition of inhibition in visual cortex, *Nature Neuroscience* 16, 1068–1076. doi:10.1038/nn.3446.

[2] Y. N. Billeh et al. (2020), Systematic Integration of Structural and Functional Data into Multi-scale Models of Mouse Primary Visual Cortex, *Neuron* 106, 388–403.e18; Fig. 4, Methods e10–e11. doi:10.1016/j.neuron.2020.01.040.

[3] L. Campagnola et al. (2022), Local connectivity and synaptic dynamics in mouse and human neocortex, *Science* 375, eabj5861. doi:10.1126/science.abj5861.

[4] G. Hahn et al. (2022), Rate and oscillatory switching dynamics of a multilayer visual microcircuit model, *eLife* 11, e77594; Fig. 5E–H. doi:10.7554/eLife.77594.